



A Nairobi train station at night. Wikimedia Commons

Power cuts are the new normal in Kenya – what went wrong and how to fix it

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Millions of Kenyan households and businesses have been subjected to interruptions of electricity supply since late 2024 owing to production shortfalls. President William Ruto acknowledged this, [explaining](#) that “daily load-shedding” had become necessary and that power would be switched off in some areas between 5pm and 10pm to stabilise the national grid.

Until now, Kenya’s electricity supply has been mostly adequate to meet supply. However, there were multiple nationwide blackouts between [2020](#) and 2024. These disruptions were due to technical failures rather than unmet demand.

The uncomfortable truth is that Kenya’s demand surge is testing the limits of what grid engineers call “[firm and operationally available capacity](#)”. This is what can be counted on when the evening peak demand rises sharply, stretching the system’s ability to maintain frequency and voltage within limits.

By the end of January 2026, the published system peak was 2,439.06 MW compared to firm capacity of 2,495 MW. There was a narrow reserve margin of only 2.3%. This peak was recorded on 4 December 2025, and was framed by Kenya Power itself as a [historic high](#).

Kenya has a reserve of nearly 800 MW on paper, but only about 56 MW of breathing room on firm capacity. This is a razor-thin margin for a system that must ride over:

- transmission constraints such as transformer overloads due to unexpected demand spikes and equipment failure
- inadequate generation forces for dispatchable baseload, from post-sunset loss of solar output of 514 MW and at times wind of 436 MW with low capacity factors

- limited flexibility to support timely ramping (how fast the rest of the system must move up or down when a generation unit trips).

My research focus is power market reforms, regulation and utility performance – including Kenya's. My assessment is that Kenya's power sector is not short of renewable energy resources to exploit. It is short of capital and a well-planned procurement pipeline of investments in new power plants and grid resilience.

Policy makers have to do more to keep up with an economy whose peak demand now resets with unsettling frequency, affecting businesses and home users.

Kenya's optimum outcome is not simply higher installed megawatt capacity. It is the combined effect of:

- sufficient energy capacity
- the system's capacity to meet fluctuating demand, changes in generation output and unexpected outages
- ability to operate, refurbish and maintain the grid network to meet set technical regulatory standards.

How did supply fall behind demand?

Three structural drivers explain the current crisis.

First, no new interconnected power plants have been commissioned during the past four years. Kenya's new capacity pipeline was constrained by a moratorium on new plants imposed in 2021. The moratorium was only lifted in December 2025 by the National Assembly, reopening the door to new procurement via competitive auctions.

Second, peak demand growth accelerated over the same time period. In February 2025, for instance, peak demand grew by the largest margin in five years. This growth was driven mainly by industrial and commercial users, a growing fleet of electric vehicles, new data centres, and an aggressive domestic power connectivity programme.

The utility surpassed 10 million customers with over 401,848 new connections in the year to 30 June 2025. This resurgence translates into a growth in sales to 11,403 GWh in just one financial year, 2024/25. The result was that a planning problem became an operational one. The mass connectivity programme stepped up over the past eight years is a triumph as the country rushes to achieve universal electrification goals. But it is also the core demand-side force compressing reserve margins.

The third factor that's affected the power network is that industrial and commercial consumers are increasingly financing their own supply. Instead of waiting for grid reliability to improve, firms have been building their own dedicated power plants. By June 2025, so-called captive (self-consumption) capacity reached 603.8 MW (about 15.72% of total installed capacity), dominated by captive solar PV and bioenergy.

While these are cheaper and more reliable sources, they are not failure-free and also serve to mask the growing national deficit.

Furthermore, this trend complicates system planning because Kenya Power's revenue base and load profile become less predictable, leading to system imbalances and frequent outages.

What's behind the instability of Kenya's electricity grid?

Kenya's energy mix is renewables-led. Renewable energy stands at 80% of the energy mix and has been steadily rising over the last 10 years.

The largest technology shares are: geothermal 943 MW (25.92%), hydro 872.5 MW (23.9%), solar 514.1 MW (14.1%), wind 436 MW (11.9%), and bioenergy 163.8 MW. The country also imports electricity from Ethiopia and Uganda, accounting for 10.6% of the total.

This picture shows why system flexibility and network reliability are key. When solar and wind power aren't available, the system must turn to geothermal, hydro and thermal while maintaining reserves.

With firm capacity only modestly above the latest peak, even a single contingency can force controlled load-shedding to preserve system integrity.

Kenya's grid instability is not one problem, however. Network reliability is undermined by system leakages from unbilled or stolen energy. In 2025, average annual losses amounted to 23.36% – far above the regulator's allowable benchmark of 17.5%. Reliability is improving, but still a far cry from best practice.

Another major factor is inadequate transmission infrastructure, primarily its high-voltage transmission lines. This means that Kenya also needs to massively invest in expanding its transmission system. Indeed, the power transmission monopoly – Ketraco – warns in its 2025-2044 master plan that keeping up with demand growth requires a multi-billion-dollar buildout. It points to an estimated financing gap of roughly US\$4.38 billion across planned transmission investments.

What's needed

Four options stand out for consideration.

The first is rebuilding the pipeline of new power plants. The quickest reliability gains will come from adding new low-carbon capacity from geothermal rehabilitation and new gas units. Policymakers must also ensure adequate extra generation capacity to provide power within seconds or minutes to cover a likely generation failure or demand spike.

Second, the system needs modern flexibility tools, such as battery storage, gas and imports. This is because storage and grid-stability investments can improve system flexibility and reduce the need for load-shedding when supply from renewables dips during peak demand.

Third, private capital participation is unavoidable if the grid is to stay ahead of demand. The most concrete step so far is the transmission monopoly's US\$311 million (KES 40.4 billion) public-private partnership signed in December 2025 with Africa50 and Power Grid Corporation of India.

Finally, stability depends on addressing system losses. This can be achieved by scaling up smart metering, restructuring distribution lines, and reducing vandalism and illegal connections. This can translate into added capacity.

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